



From Input to Output: The LLM Pipeline

Demystifying the inner workings of Large Language Models

Exploring the journey of text, from prompt to completion

Understanding core components and their interactions

1. Tokenization: Breaking Down the Input

Text-to-Tokens

Convert raw text into numerical tokens

Vocabulary Map

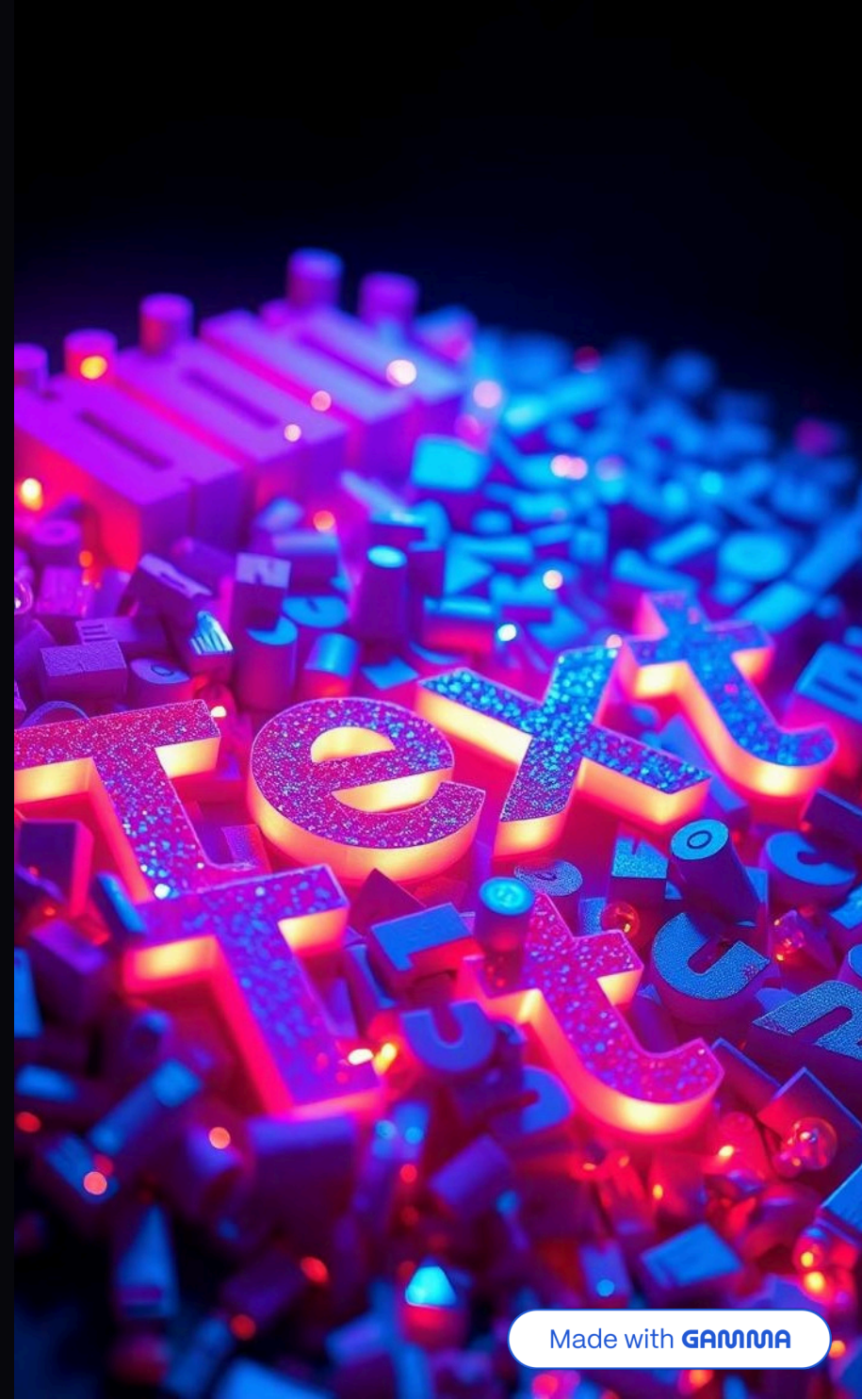
Words/subwords mapped to unique IDs

Methods

- Byte Pair Encoding (BPE)
- WordPiece

Example

"The cat sat on the mat." → [42,1209,56,32,887,13]



2. Embedding: Representing Tokens as Vectors

Vector Transformation

Tokens converted to continuous embeddings

Dimensions: 512-2048 typical range

Semantic Meaning

Close vectors = related tokens

Embedding matrices are learned lookup tables

Examples & Tools

Token ID 42 ("The") → vector

Popular pre-trained embeddings:
GloVe, word2vec, fastText

3. Transformer Layers: Processing the Embeddings

Self-Attention

Captures relationships between tokens

Multi-Head Attention

Parallel attention layers for diverse context

Feed-Forward Networks

Non-linear transformations of attention outputs

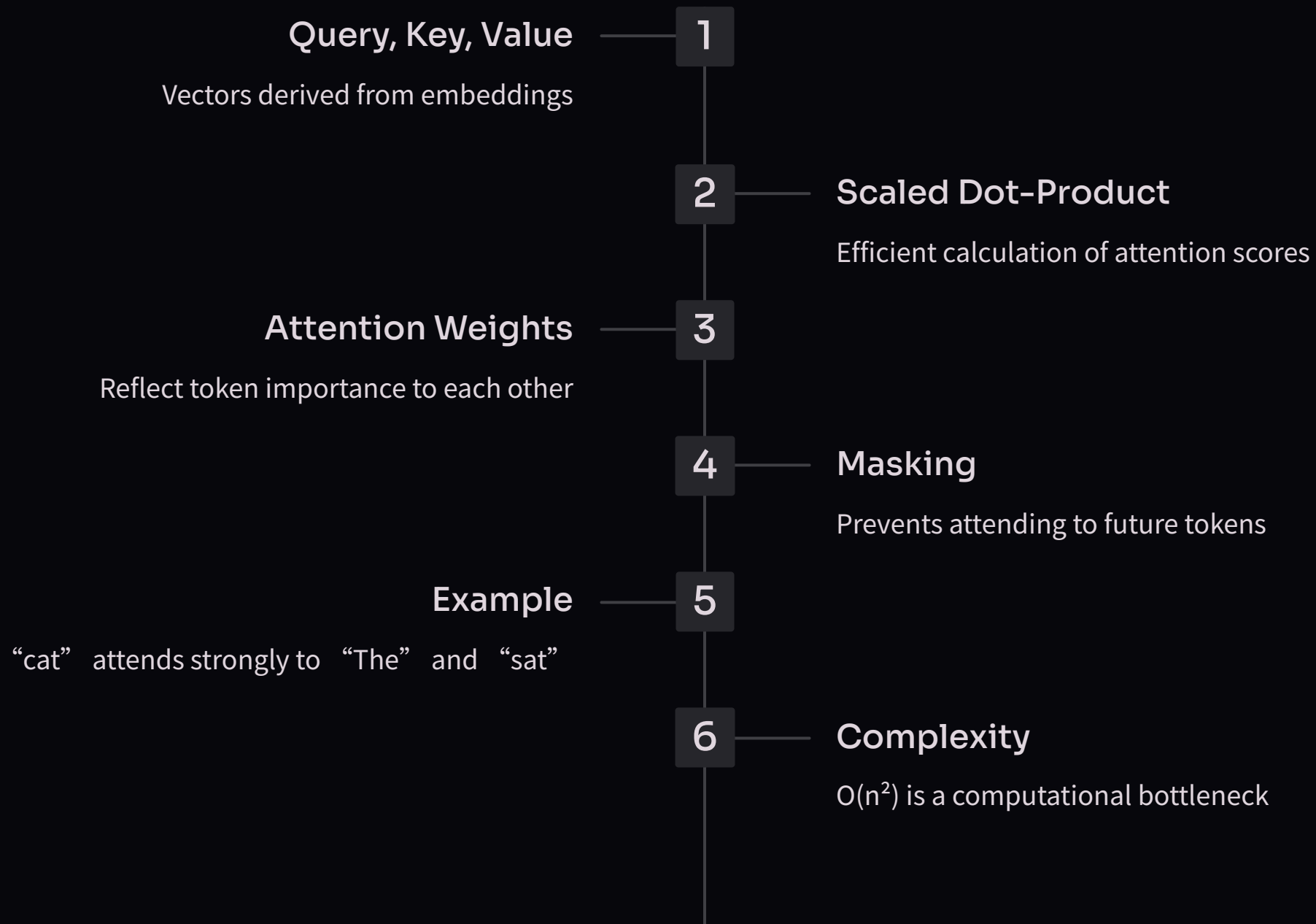
Other Components

Residual connections, layer normalization, positional encoding

Architectures

BERT uses encoder-only, GPT uses decoder-only

4. Self-Attention: Understanding Token Relationships



5. Decoding: Generating the Output Tokens

1

Autoregressive Generation

Predict next token using previous context

2

Softmax Layer

Outputs token probabilities

3

Sampling Techniques

- Greedy decoding
- Beam search
- Top-k sampling
- Nucleus sampling

4

Control

Temperature adjusts randomness of output

5

Example

Next word after “The cat sat on” : “mat” highest prob

6. Post-Processing: Refining the Output

Detokenization

Tokens converted back into natural language

Clean-Up

Remove special tokens like [CLS], [SEP], [PAD]

Text Fixes

Apply grammar, capitalization, punctuation corrections

Filtering

Remove inappropriate or harmful content



7. Challenges and Limitations

Resource Intensive

High computational cost for training and inference

Data Bias

Models may amplify training data biases

Interpretability

Hard to explain why predictions occur

Hallucinations

Generation of false or nonsensical content

Example: GPT-3 trained on 45TB of diverse text data



Conclusion: The Power and Complexity of LLMs

Powerful NLP Tools

Enable advanced language understanding and generation

Responsible Use

Understanding components key for ethical applications

Ongoing Research

Address limitations and enhance capabilities

Future Outlook

More efficient, accurate, and ethical models ahead